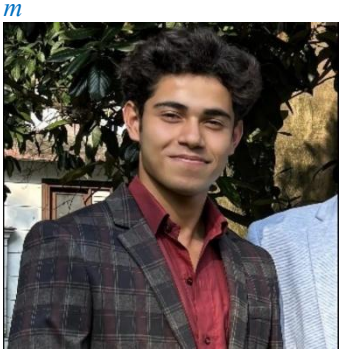
# PROJECT AND TEAM INFORMATION

## Project Title

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| **NEXUS** - AI Powered Distributed Data Processing & Computation Orchestration Platform. |

## Student / Team Information

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|---|---|
| ***Team Name:***<br>***Team ID:*** | ***Avengers***<br>***OS-DBMS-V-2026-T099*** |
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## Motivation

Modern organizations generate a large amount of data, but analyzing this data can be difficult for users who do not have good knowledge of SQL and database systems. Even when databases are available, users often need to write complex SQL queries manually. Another problem is that different queries require different amounts of CPU, memory and processing time. Large analytical queries can consume many resources, while simple queries may require very little. Traditional systems do not always connect query processing with resource management. NEXUS is proposed to solve these problems by combining Artificial Intelligence with DBMS, Operating Systems and Distributed Systems. Users can upload a dataset and ask questions about it using normal language instead of writing SQL manually.

For example, a user can ask: "Find the top ten cities by revenue."

The AI converts this request into an SQL query. The query is then checked for correctness and safety, optimized and sent to the job scheduler. The scheduler assigns the job to an available worker. This makes data analysis easier for normal users while also demonstrating important computer science concepts such as database management, process scheduling, concurrency, resource allocation and distributed execution.

## State of the Art / Current solution

Currently, data analysis is mainly performed using database management systems, SQL tools, dash-boards and data-analysis software. These systems are powerful, but users often need knowledge of SQL and database structures to perform advanced analysis. Some modern AI tools can convert natural-language questions into SQL or provide summaries of data. However, these AI tools are generally focused on the AI part and may not provide complete control over query validation, optimization, scheduling and computational resources. Similarly, distributed computing platforms can execute workloads using multiple machines or workers, but they are generally designed for technical users and require knowledge of data-processing systems. NEXUS combines these different capabilities into one platform. It provides a natural-language inter-face using AI, validates and optimizes generated queries, schedules computational jobs and distributes them among workers. Therefore, the proposed system connects AI-based data interaction with database management and operating-system-based resource management in a single application.

## Project Goals and Milestones

The main goal of NEXUS is to develop a working web application that allows users to upload datasets, analyze them using natural language and monitor the execution of their analytical jobs.

**I. Project Goals :**

1. Develop a web interface for users.
2. Implement user authentication and role-based access.
3. Allow users to upload and manage datasets.
4. Allow users to ask questions using natural language.
5. Convert natural-language requests into SQL using AI.
6. Validate generated SQL before execution.
7. Optimize queries for better performance.
8. Implement a job queue and scheduling system.
9. Execute jobs using multiple worker processes.
10. Monitor CPU, memory and worker usage.
11. Store query history and execution logs.
12. Provide visual results and AI-generated explanations.

**II. Initial Milestones :**

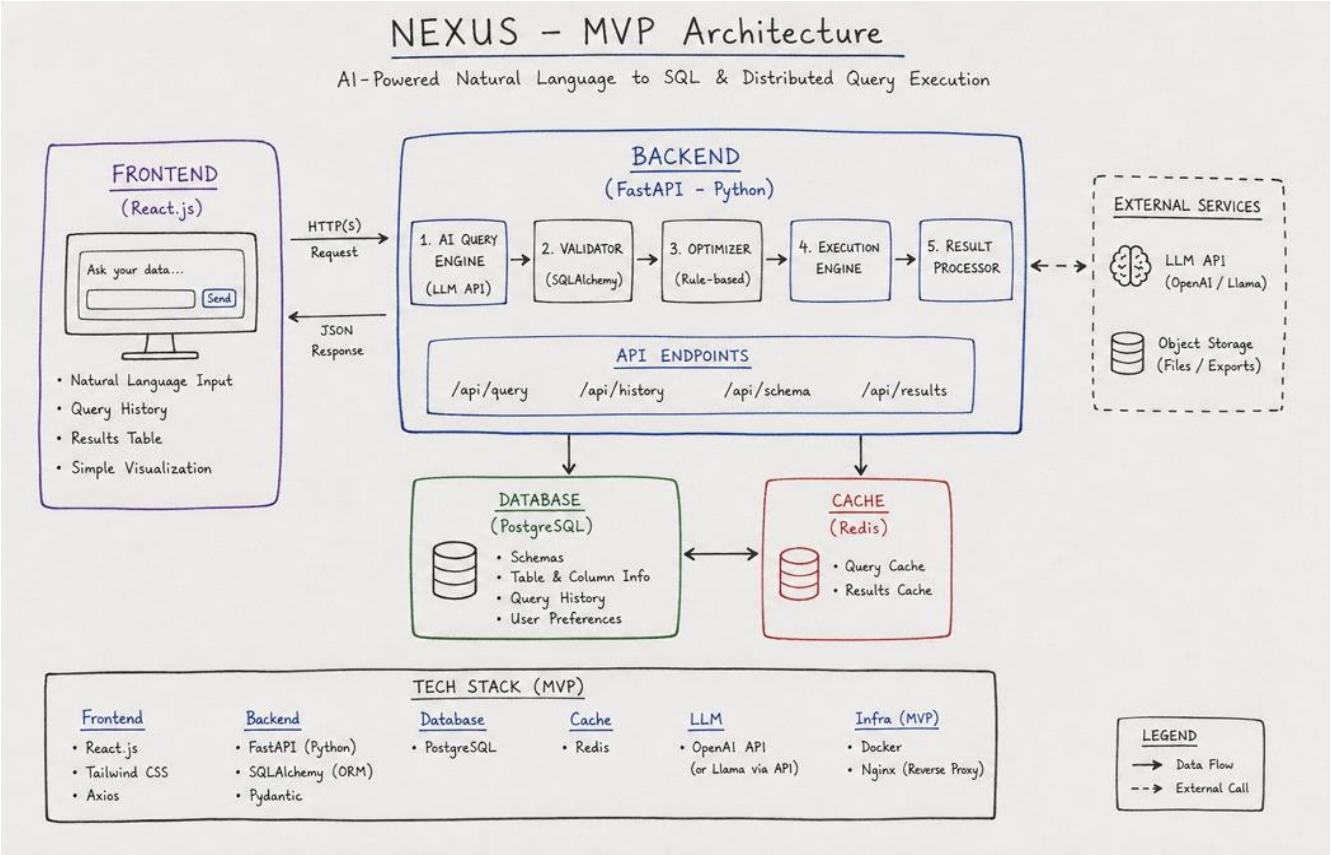
1. Requirement analysis and system architecture.
2. Design and implementation of the PostgreSQL database.
3. Development of frontend, authentication and dataset management.
4. Development of AI-based natural-language-to-SQL system.
5. Implementation of SQL validation and query optimization.
6. Development of job scheduling and worker management.
7. Implementation of monitoring, logging and result visualization.
8. Integration, testing and final deployment.

These milestones cover the major objectives described for NEXUS.

## Project Approach

NEXUS will be developed as a full-stack web application using a modular architecture. The frontend will provide the user interface for authentication, dataset upload, natural-language queries, result visualization and job monitoring. React/Next.js will be used for the frontend. The backend will be developed using Python FastAPI. It will manage communication between the frontend, AI system, database and job scheduler. PostgreSQL will be used as the main relational database. It will store users, datasets, queries, jobs, workers and execution information. The AI component will use an LLM API or local LLM to understand natural-language questions and generate SQL queries using database schema information. The generated query will first pass through a validation layer that checks its syntax, permissions, complexity and safety. After validation, the Query Optimizer will find an efficient execution strategy. The query will then be converted into a job and placed in the scheduler queue. The scheduler will use concepts such as FCFS, Round Robin, Priority Scheduling, Shortest Job First and Resource-Aware Scheduling. Worker processes will execute the jobs concurrently. Redis or a custom scheduler can be used for task management, while Docker can be used to run different services and worker nodes. Git/GitHub will be used for version control.

## System Architecture (High Level Diagram)




## Project Outcome / Deliverables

The final outcome of the project will be a working web application called NEXUS.

The major deliverables will include:

1. A web-based user interface.
2. User authentication and role-based access control.
3. Dataset upload and management.
4. Automatic dataset schema detection.
5. Natural-language query interface.
6. AI-based SQL generation.
7. SQL validation and safety checking.
8. Query optimization.
9. Job queue and scheduling system.
10. Distributed worker execution.

11. Query history and execution logs.

12. System monitoring dashboard.

13. Query result tables and visualizations.

14. AI-generated explanations and insights.

The completed system will demonstrate the practical integration of DBMS, Operating Systems, Distributed Systems and Artificial Intelligence.

## Assumptions

The project assumes that users will mainly upload structured datasets such as CSV files. PostgreSQL will be available as the primary database, and an LLM API or locally hosted AI model will be available for natural-language-to-SQL generation. The initial system will use a limited number of worker processes or containers. It is also assumed that users have the required permissions to access the datasets they query. All AI-generated SQL queries will pass through validation and authorization before execution.

## References

1. [1] P. A. Prakash, S.Shakthi, M. P. Sri Santhiya, A. V. Shreeja, S. Subash, and C. Eyamini, "AI-Powered Transformer Framework for Intelligent Natural Language to SQL Query Generation," in Proc. 2026 Int. Conf. Smart Futuristic Technology (ICSFT), 2026, doi: 10.1109/ICSFT66733.2026.11507944.
2. [2] S. Bagui, M. Malagutti, R. Morelli, and M. Tamascelli, "Large Language Models in Database Management System Optimization: A Survey," *ACM Trans. Intell. Syst. Technol., vol.*17,no. 5, Art. no. 119, 2026, doi: 10.1145/3829078.